

Learning goals: Basics of shell scripting

NOTE: Please do all your work in a new folder called `lab9`. Save the script for each exercise separately as `question1.sh`, `question2.sh` and so on. At the end of the lab, please upload your scripts to your github repository.

EXERCISES

- (1) Using `echo` to print a multi-line message inside a script: The `-e` option to `echo` command enables the interpretation of backslash escapes. `\n` represents a newline character, it leads to subsequent text to be printed on a new line. `\t` represents a tab character, causing subsequent text to be indented with a tab. Try the following command to understand how this option works-

```
echo -e "This is line number 1.\n This is line number 2.\n\t This is line number 3 indented with a tab character"
```

Inside a shell script, use the `echo` command to output the following text in the given format below. Note that each item below is indented with a tab character.

Tips for programming:

1. Master the fundamentals
2. Learn by doing
3. Understand the "why"
4. Practice regularly
5. Debug effectively
6. Read and analyse code
7. Break down problems
8. Utilize resources
9. Don't fear errors
10. Start small and iterate

- (2) Write a shell script to print your name and age as "My name is <so-and-so>, and I am <so-many> years old."
- (3) Write a shell script that creates an array called `colors` containing the names of your favorite colors. Print the entire array.
- (4) There are four 'special' variables that a shell uses for its functioning. You have already seen one of these- `$?`. The other three are

`$0` represents the name of the script itself

`$#` represents the number of arguments passed to the script

`$@` represents all the arguments passed to the script

Write a shell script with 4 `echo` commands to display the value of each of these variables

- (5) By extension the shell interprets arguments to any script as `$1`, `$2` etc. See the example below to understand how this works. Create the following shell script:

```
#!/bin/bash
```

```
echo "My name is $1"
```

Save the above script as `name.sh`. Then make it executable using the `chmod` command, and execute it as follows: `./name.sh <yourname>`. The output will be (in my case) `My name is Jayashree`. `$1` is the first argument to the shell script which is automatically read in and used in the execution of the script.

- (i) Write a script to print your name as well as the MSc program you are in by using `$1` and `$2`.

- (ii) Now add the 4 **echo** lines from your previous question to this script and execute the script. What are the values of the special variables? Take a snapshot of the output and upload this png/jpeg file with a name with this question number.
- (6) Write a shell script to do the following, all in one script:
- Print the value of HOME variable
 - Calculate the following: $23934/44343$ using **bc** command and **EOF** method
 - List files in your HOME that start with 'D'
 - Filter the line(s) in /etc/passwd that have your username
- (7) Write a shell script that calculates the energy-mass equivalence ($Energy = mass * speed^2$) of an object with mass of 1kg moving with the speed of light (what is this value?). Note: It's best to invoke **bc** command with multiple expressions using **EOF** approach.
- (8) Write a shell script that directs the output of **ls** command to a **grep** command to list files with extension **.csv**. Make sure to copy or download the two data files you have used in this course before to this directory first (these are BrainCancer.csv and Heart.csv).